

INDR-100 HOMEWORK-1

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Question-1

1) (25 pts) **EcoRide Scooter Refurbishment Facility:** EcoRide is a startup that buys used electric scooters, refurbishes them, and resells them. Currently, the facility suffers from a poorly planned layout, causing workers to walk back and forth unnecessarily, materials to be transported inefficiently, and long waiting times due to bottlenecks. As the facility manager,

a) (12 pts) The 12 process steps that take place in the facility are listed below in random order. Arrange these steps in a logical production flow. Then, for each step, write the name and draw the symbol of the appropriate Flow Process Chart activity (Operation, Transportation, Inspection, Delay, or Storage).

1	4	The refurbished battery and motor are mounted onto the frame	Storage	Δ
2	6	The fully assembled scooter is examined in the test room according to riding and safety standards.	Inspection	□
3	7	The battery and motor are separated from the frame	Operation	○
4	1	The scooter frame remains stationary in front of the station until the next assembly worker becomes available.	Operation	○
5	10	A scooter is taken from the warehouse and transported to the pre-registration desk by a cart.	Transportation	→
6	12	The battery and motor are sent to the testing station, where they are connected to measurement devices and checked against quality standards.	Inspection	□
7	8	Once the test outcome is recorded, the battery and motor are placed on hold alongside the already-queued frame until a worker becomes available.	Transportation	→
8	11	The frame and the battery/motor set are transported to the relevant assembly table.	Transportation	→
9	3	After the intake processing steps are completed at the desk, the registered scooter is transferred to the disassembly station via a conveyor belt.	Transportation	→
10	9	Finished scooters are held in the storage area until they are picked up for shipment.	Operation	○
11	2	Scooters that have completed and passed all tests are boxed.	Inspection	□
12	5	Used scooters arriving from outside are held in the main warehouse under authorization until they are ready to be processed.	Storage	Δ

Department Assignment

Room 1: _____	Room 4: _____
Room 2: _____	Room 5: _____
Room 3: _____	Room 6: _____

- b) (6 pts) Considering the 12-step flow you arranged/sorted in part (a), place the six departments below into the appropriate areas on the blank layout provided. Your placement should minimize transportation distances, prevent backtracking, and create a sequential, smooth production line. Departments: Raw Material Storage, Pre-registration & Disassembly Desk, Testing Station, Assembly Station, Final Check Room, Finished Goods Storage

Solution:

We want to optimize the refurbishment process and minimize material handling. I assigned the rooms on the sequential flow derived in part (a)

- **Room 1: Pre-registration & Disassembly Desk** - Located at the entrance for first processing
- **Room 2: Testing Station** – Next to the disassembly to verify parts
- **Room 4: Raw Material Storage** – Adjacent to assembly to feed new parts
- **Room 5: Assembly Station** – Centrally located to receive refurbished parts from room 2 and new parts from room 4
- **Room 3: Final Check Room** – Next to assembly before final storage
- **Room 6: Finished Goods Storage** – Located at the exit for shipping

Interpretation:

I ensure this assignment a unidirectional flow. Assembly (Room 5), Testing (Room 2), and Final check (Room 3). I eliminate unnecessary movement and prevent backtracking to increase efficiency.

- c) (7 pts) On your Suggested Layout in part (b), draw the route that materials follow using arrows. Then, write the number of each of the 12 steps inside the correct Flow Process Chart symbol, and place these symbols on the relevant departments or routes on your layout sketch. Hint: The flow should be placed on the layout, where the departments are placed, as in Lecture 10, slide 28.

Solution:

I create this workflow is mapped onto the facility layout, from part a to directional arrows:

- **Entrance-> Room 1:** Step 4(Δ), Step 6(□), and Step 7(○) occur here.
- **Room 1-> Room 1:** Step 1(○) – Frame washing is done withing the disassembly area
- **Room 1-> Room 2:** Step 10(→) change the place of components for testing
- **Room 2:** Step 12(□) – Check quality standards for battery/motor.
- **Room 2-> Room 5:** Step 7(→) change the place of functional parts of the assembly.
- **Room 4-> Room 5:** Step 3-11(→) bring the new parts and the cleaned frame to the assembly point
- **Room 5:** Step 9(○) -Full reassembly of scooter.
- **Room 5-> Room 3:** Step 2(→ to □) change the place of the unit to final inspector.
- **Room 3-> Room 6:** Step 5(Δ) – Final storage before exiting via Shipping.

Interpretation:

I use this visualization to confirm that the material flow logically from the top-left Assembly towards the bottom_right(Exit). Frame's path and components converge at the Assembly Station(Room 5) without any crossing paths, that decreases congestion and reduces the risk of process delay.

3) (25 pts) EcoRide Scooter Refurbishment Facility (cont.): For the EcoRide facility introduced in the previous question, answer the following.

a) Utilization and Efficiency Calculations

i) Design Capacity: 240 units/day

ii) Effective Capacity: 200 units/day

iii) Actual Output: 180 units/day

Formulas:

Calculations:

$$Utilization = \frac{\text{Actual Output}}{\text{Design Capacity}} \times 100$$

$$Utilization = \frac{180}{240} \times 100 = \mathbf{75\%}$$

$$Efficiency = \frac{\text{Actual Output}}{\text{Effective Capacity}} \times 100$$

$$Efficiency = \frac{180}{200} \times 100 = \mathbf{90\%}$$

Interpretation: The facility is operating at 75% of its maximum potential(design) and 90% of its planned(effective) performance.

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